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Evaluating AI-Assisted Customer Verification for DNA Synthesis Screening

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2 ABSTRACT

3 Legitimacy screening, the process of verifying the identity and purpose of customers ordering
4 synthetic nucleic acids, is a primary safeguard against the misuse of synthetic biology, but its cost
5 and labor intensity remain as obstacles to widespread adoption. To evaluate whether AI tools can
6 accelerate this process, we tested five large language models on five verification tasks using 41
7 simulated customer profiles of life sciences researchers from around the world. We compared AI
8 performance against an expert human baseline on flag accuracy, source quality, source fidelity,
9 and cost.

10 The best-performing model (Gemini 2.5 Pro with specialized tools) passed 89.8% of evaluation
11 tests, compared to 79.5% for the human baseline, at a total per-customer cost of \$1.18 versus
12 \$14.04—roughly a 10-fold reduction. For tasks AI could perform without human intervention, costs
13 averaged \$0.23 per customer, roughly 50 times cheaper than human screening. These results
14 support piloting AI-assisted legitimacy screening at DNA synthesis providers, with AI systems
15 handling information gathering and human reviewers retaining authority over order fulfillment
16 decisions.

17 **Keywords:** biosecurity, DNA synthesis screening, large language models, legitimacy screening, customer verification, know your
18 customer, artificial intelligence, synthetic biology

1 INTRODUCTION

19 Synthetic nucleic acids are unique among commercially available laboratory reagents: they can be used to
20 directly create pathogens through simple laboratory techniques, without requiring a live pathogen culture.
21 Most uses of synthetic nucleic acids benefit health and biosecurity, such as through vaccine development,

diagnostics, and biotechnology research. However, a bad actor could purchase commercially available synthetic nucleic acids to create dangerous pathogens. The sequences that enable this are called “sequences of concern” (SOCs). Screening customers who order SOCs is therefore a primary safeguard against misuse.

As nucleic acid synthesis becomes more accessible, strengthening these safeguards becomes more important. Organizations like the International Gene Synthesis Consortium (IGSC) have committed to two forms of screening: sequence screening (checking ordered DNA against databases of regulated pathogens) and legitimacy screening (verifying that customers have legitimate reasons to order flagged sequences). However, widespread adoption remains limited (Crawford et al., 2024). The synthesis industry is highly competitive and robust customer screening can present a conflict: legitimacy screening—particularly the follow-up verification triggered by risky sequences—requires substantial manual work (Carter and Friedman, 2015; Alexanian and Carter, 2024). For gene-length orders, recent work calculates a per-order cost of £0.95 for legitimacy screening compared to £0.10 for sequence screening—almost an order of magnitude difference (Fady et al., 2025a).

Currently, no jurisdiction requires providers of synthetic nucleic acids to screen customers. While anti-terrorism laws create implicit obligations, these are neither standardized across countries nor grounded in a shared risk framework. Screening practices therefore vary substantially across providers, creating uneven security guarantees against self-amplifying biological threats which easily cross borders.

Legitimacy screening typically involves two phases: onboarding and follow-up. Onboarding checks verify basic customer information: name checks against watchlists, institutional affiliation, email domain ownership, and stated research purpose. Sequence screening compares ordered DNA against databases of regulated pathogens to identify sequences of concern. When a sequence of concern is flagged, follow-up screening determines whether the customer has a legitimate need for it. This phase requires gathering information from publications, patents, and institutional records, plus requesting documentation from customers such as grant awards or biosafety approvals. The information-gathering step is time-consuming but largely mechanical: it involves searching public records and rarely requires judgment. This makes it a natural target for AI assistance.

The aim of this study was to evaluate whether AI tools can accelerate information gathering as part of legitimacy screening. We evaluated five large language models with web search capabilities and with additional specialized tools on five verification tasks relevant to follow-up screening: institutional affiliation, institution type, email domain ownership, sanctions screening, and background work search. We found that AI-assisted screening reduces total per-customer costs roughly 10-fold compared to manual screening (\$1.18 versus \$14.04), while matching or exceeding human accuracy. For tasks AI could perform without human intervention, costs averaged \$0.23 per customer—roughly 50 times cheaper than human screening.

2 METHODS

2.1 Task Definition

We evaluate AI and human screeners on five verification tasks relevant to customer follow-up screening:

1. **Institutional affiliation verification:** Confirm the customer is currently affiliated with their claimed institution.
2. **Institution type verification:** Confirm the institution is a legitimate life sciences research organization or company.

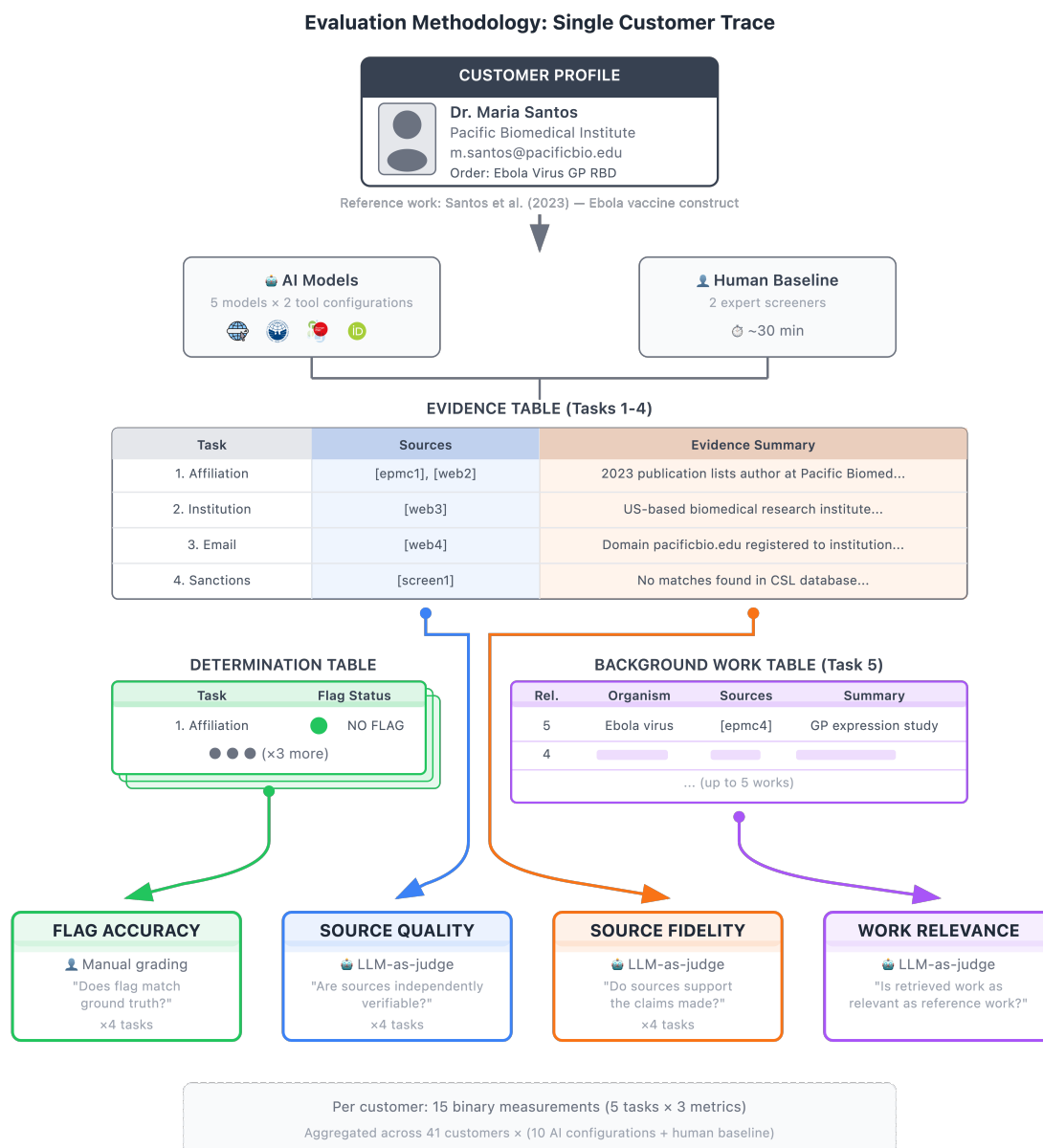


Figure 1. Evaluation methodology for AI-assisted legitimacy screening. Each customer profile includes a simulated order for a sequence of concern plus a reference work (publication, patent or news article). Screeners produce structured outputs evaluated on three binary metrics per task: source quality, source fidelity, and flag accuracy / work relevance. The example shown uses a fictional customer profile for illustration purposes.

3. **Email domain verification:** Confirm the customer's email domain belongs to their claimed institution.
4. **Sanctions screening:** Check whether the customer or their institution appears on export control or sanctions lists.
5. **Relevant work search:** Find publications, patents, or other work from the customer or their institution related to the ordered sequence.

For tasks 1–4, screeners assign a determination: flag (concern identified or information not found), no flag (verified without concern), or undetermined (insufficient evidence to decide). For task 5, screeners report relevant work without making a flag determination.

These tasks were selected because they can often be resolved using publicly available information, they appear in existing screening guidance (Alexanian and Carter, 2024), and they represent the information-gathering phase of legitimacy screening rather than the judgment-intensive decision phase.

After information gathering, a human reviewer decides whether to ship the order, reject it, or request additional documentation from the customer. We evaluate screener performance on the information-gathering tasks only; the order fulfillment decision itself is not evaluated, but its time and cost are included in our workflow estimates.

2.2 Dataset

Our primary evaluation dataset comprises 41 customer profiles, each paired with a simulated order for a sequence of concern. Each profile includes the customer's name, institutional affiliation, email address, and a reference work (publication, patent, or news-reported activity) through which we identified the customer. All profiles were simulated: no real customer orders were used, and no synthetic nucleic acids were synthesized or shipped as part of this study.

Flag accuracy was manually graded on this dataset, and automated metrics—source quality, source fidelity, and work relevance—were computed for all screeners. We also collected an extended dataset of 134 profiles on which AI models, but not human screeners, were evaluated using the automated metrics (see **Supplementary Material**, Appendix F).

To simulate realistic screening scenarios, we compiled profiles of researchers identified through publications and patents—some involving work on controlled pathogens, others on sequences that would not trigger screening concerns. We sampled across geographic regions and institutional types, and drew from the U.S. Consolidated Screening List (CSL) to include profiles that should trigger sanctions-related flags.

2.2.1 Customer Types

We constructed four groups of customer profiles:

Academic users of SOC: Life science researchers from academic institutions with documented laboratory work on SOC.

Industry users of SOC: Researchers at life science companies with documented work on SOC, performed either by the customer or by their institution.

Customers in the U.S. Consolidated Screening List: Researchers at academic institutions appearing on the CSL, with documented work on SOC.

99 **General life science researchers:** Researchers from academia and industry with documented laboratory
100 work on non-SOC sequences such as antibodies (scFv fragments), CAR-T constructs, protein engineering
101 targets, and diagnostic antigens.

102 All profiles in our dataset—including general life science researchers identified through non-SOC work—
103 were assigned simulated orders from a standardized list of 22 SOC proteins. We selected proteins that
104 would be reasonable to order for vaccine or therapeutics research while still being categorized as sequences
105 of concern (see **Supplementary Material**, Appendix A).

106 Screeners therefore encountered cases where researchers appeared to order SOC proteins without relevant
107 background—a mismatch they were expected to identify and note.

108 2.2.2 Data Collection

109 **Academic researchers:** We searched Google Scholar using each sequence name as the primary search
110 term. To promote geographic diversity, we appended personal names sampled from a random name
111 generator weighted toward common names in different world regions.

112 We selected publications meeting three criteria: (1) published after 2021, to increase the likelihood that
113 affiliations remain current; (2) involving laboratory work on the target organism or protein, rather than
114 purely computational or epidemiological analysis; and (3) plausibly requiring DNA synthesis, such as
115 protein expression, cloning, or vaccine construct development. We filtered for authors with a publicly
116 available email address or an institutional email domain identifiable through Google Scholar, ORCID, or
117 similar research profiles.

118 **Industry researchers:** We used two search strategies. For patent-based collection, we queried
119 PatentScope using organism names, prioritizing patentees from outside the US and EU to encourage
120 geographic diversity. We excluded large pharmaceutical firms, focusing instead on smaller companies that
121 are harder to verify and more likely to require individual legitimacy screening.

122 For news-based collection, we searched Google for articles combining organism names with terms
123 like “vaccine,” “diagnostics,” and “startup,” then identified researchers in laboratory roles at the resulting
124 companies via LinkedIn.

125 **CSL-affiliated academics:** We manually reviewed the U.S. Consolidated Screening List for institutions
126 conducting biological research—primarily academic institutions in China, Russia, and Iran. We then
127 searched Google Scholar using SOC organism names combined with institution names to find publications
128 by authors at those institutions.

129 2.3 Screeners

130 We compared human and AI screeners on the information-gathering component of legitimacy screening.

131 2.3.1 AI Models

132 We tested five large language models with web search capabilities: Claude Sonnet 4 (Anthropic), Gemini
133 2.5 Pro (Google), Grok 4 (xAI), GLM 4.6 (Zhipu AI), and Minimax M2 (MiniMax). All models were
134 released before November 1, 2025. We selected the latest available models from major commercial
135 providers and included GLM 4.6 and Minimax M2 as competitive open-source alternatives based on public
136 benchmark performance and LMArena rankings. Several models were excluded because they frequently
137 refused to complete the screening task, citing biosecurity concerns, including OpenAI’s GPT-5, o1, and o3,
138 and Anthropic’s Claude 4.5 Sonnet.

Table 1. Dataset composition by customer category. Numbers outside parentheses are the 41-profile primary dataset; numbers in parentheses are the extended 134-profile dataset. Region is determined by institution location. Institutional email indicates profiles with a business, academic, or government email domain.

	Academic SOC Users	Industry SOC Users	Life Sci. Researchers	CSL Academics	Total
Profiles	16 (56)	11 (24)	7 (29)	7 (25)	41 (134)
Region					
US	2 (6)	4 (8)	0 (1)	0 (0)	6 (15)
Europe + Oceania	5 (18)	2 (6)	0 (4)	0 (6)	7 (34)
China	2 (8)	3 (6)	6 (19)	1 (5)	12 (38)
Other	7 (24)	2 (4)	1 (5)	6 (14)	16 (47)
Inst. email	15 (49)	10 (22)	5 (18)	3 (19)	33 (108)

Each model was tested under two conditions:

Web search only: Models had access to web search via Tavily, a search API designed for AI applications, and no other tools.

Web search plus specialized tools: Models had access to web search plus four additional tools:

- *Consolidated Screening List API:* Searches the U.S. government's consolidated list of sanctioned entities and restricted persons.
- *Europe PMC:* Searches Europe PubMed Central for scientific articles by author, institution, topic, or ORCID identifier.
- *ORCID profile:* Retrieves researcher profile information including affiliations, employment history, and recent publications.
- *ORCID works search:* Searches a researcher's full ORCID publication list by keywords.

Both conditions used identical screening prompts (see **Supplementary Material**, Appendix B).

2.3.2 Human Baseline

Two coauthors served as the expert human baseline. Collecting customer profiles and grading outputs were conducted by the lead author, who did not serve as a screener.

Both evaluators were familiar with the research plan but received no task-specific training. They were given an earlier version of the screening instructions (developed while optimizing prompts for model performance) and had not previously seen the profiles they evaluated. They submitted responses through a custom interface that enforced the same output format as model responses.

The interface recorded snapshots of each screener's work at approximately 5 and 30 minutes. If they submitted a final answer earlier, that submission was used as the 30-minute snapshot. Each evaluator screened 20 or 21 profiles, covering the full primary dataset. Profiles were randomly sampled with the constraint that no two shared the same reference work.

We collected the 5-minute snapshot to estimate human performance under a time budget comparable to AI models, which averaged under 2 minutes per profile. However, responses at 5 minutes were typically incomplete, and this baseline underperformed AI models by a large margin. Unless otherwise noted, "human baseline" refers to the 30-minute snapshot throughout.

2.4 Evaluation Metrics

We evaluated each of the five verification tasks on three binary metrics, yielding 15 measurements per customer profile. To scale evaluation, we used Gemini 2.5 Flash in an LLM-as-a-judge setup for all metrics except flag accuracy, which was manually graded (see **Supplementary Material**, Appendix C).

2.4.1 Verification Tasks (Tasks 1–4)

Flag accuracy: Whether the response's determination (flag, no flag, or undetermined) matches ground truth. Ground truth was established through manual review of all model and human responses. When responses disagreed, we reviewed justifications and occasionally followed cited sources to verify claims. We were not blind to which screener produced each response, but none of the screeners participating in the human baseline were involved in ground truth annotation.

Source quality: Whether cited sources meet standards for independent verification. A source is acceptable if it exists independently of the customer and has editorial oversight. Acceptable sources include government registries, peer-reviewed publications, patent filings, regulatory submissions, business registrations, and established research profile aggregators. Unacceptable sources include LinkedIn profiles, personal websites, Wikipedia, and social media sites. A response passes this metric if all cited sources are acceptable; responses citing any unacceptable source, or no sources at all, fail.

Source fidelity: Whether factual claims are directly supported by cited sources. A claim fails if the cited source contradicts it, no cited source provides relevant information, or the claim presents speculation as established fact. Claims reporting absence of evidence (e.g., “no matching records found”) passed by default, as negative claims are inherently difficult to source. Empty responses automatically failed.

2.4.2 Background Work Task (Task 5)

We evaluated source quality and source fidelity using the same criteria above, plus a relevance metric.

Work relevance: Whether the response identifies work at least as relevant as the reference work used to include the customer in our dataset. Relevance is scored 0–5 based on three factors:

- *Proximity to customer:* Work authored by the customer directly (higher) versus produced by their institution (lower).
- *Organism proximity:* Same organism as the order (higher), closely related organism (middle), or unrelated (lower).
- *Laboratory involvement:* Hands-on experimentation (higher) versus purely computational work (lower).

A response passes if at least one retrieved source scores at or above the relevance level of the reference work.

2.5 Prompt Development

We used the same task prompts across all models and developed them through iterative refinement on 10 customer profiles. After each iteration, we reviewed outputs for common errors such as citing unrelated sources, making overly broad claims, failing to search multiple sources before concluding information was unavailable, and failing to comply with sequential instructions. We rewrote and consolidated instructions multiple times to improve task adherence (see **Supplementary Material**, Appendix B).

Evaluation prompts underwent a similar iterative process to improve alignment with human judgment. To validate the LLM-as-judge approach, one author independently re-graded a sample of 135 evaluation

outputs without access to the original model grades. Agreement reached 83%, compared to 63% expected by chance (Cohen's $\kappa = 0.63$). This falls within the “substantial agreement” range (0.61–0.80) on the Landis and Koch (1977) interpretation scale (see **Supplementary Material**, Appendix C).

2.5.1 Cost and Time Estimation

We estimate the marginal cost of screening an additional customer. Fixed costs such as prompt development, tool integration, and screener training can be amortized over a large number of cases and are excluded from our estimates.

For both workflows, we measure cost through the initial screening step: from receiving a customer profile to recording an order decision. When the decision is to request follow-up, we do not include the cost of subsequent customer communication or verification of additional information.

Human-only screening costs: We estimated costs based on time required and hourly wage for comparable roles. Using advertised salaries for customer service positions at large DNA synthesis providers, we estimate an hour of human screening at approximately \$54. This figure depends heavily on the country in which a screener is employed; due to limited salary data, we treat it as representative of industry-wide costs.

AI-aided screening costs: For information gathering, costs were based on each provider's per-token API pricing (with input and output tokens priced separately) plus the cost of each web search query requested by the model. For order fulfillment decisions, we timed the human review process and estimated costs using the same hourly rate.

3 RESULTS

3.1 Pass Rates by Screener

As Figure 2 shows, AI models outperformed the human baseline on most metrics. For every metric other than flag accuracy, all models achieved a higher pass rate than both the 5-minute and 30-minute human baselines. Even for flag accuracy, the 30-minute human baseline outperformed only two of the ten model configurations. The 5-minute human baseline performed much worse than any other screener setup, largely because responses were practically always incomplete, leading to automatic failures on evaluation metrics.

Across AI models, pass rates spanned a narrow range: the best-performing configuration (Gemini 2.5 Pro with specialized tools) passed 89.8% of tests, while the worst-performing (Gemini 2.5 Pro with web search only) passed 83.7%—notably, the same underlying model in both cases, differing only in tool access.

Access to specialized tools (Consolidated Screening List API, Europe PMC, ORCID) affected performance differently depending on the task. For verification tasks (Tasks 1–4), tool access produced a modest but consistent improvement across all metrics. The improvement was most pronounced for sanctions screening (**Supplementary Figure E3**), where the Consolidated Screening List API provided direct access to authoritative data otherwise available only through downloadable files or specialized interfaces.

For work relevance (Task 5), the pattern reversed: models with tool access scored lower than those with web search only. The drop was concentrated among industry customers (**Supplementary Figure E4**), while academic and other customer types showed little difference between configurations.

This likely reflects the fact that models with tool access performed notably fewer web searches (**Supplementary Figure I1**), making them more likely to miss patents or news articles—sources not

Table 2. Per-customer screening costs and processing times. “Information gathering” covers Tasks 1–5 only; “total cost” adds the time cost of human review of the AI-generated report. For human baselines, these phases were not separated, so only totals are reported. Human costs estimated at \$54/hour based on advertised salaries at a large DNA synthesis provider. AI costs include per-token API pricing and Tavily web search queries (\$0.08/query); other tools were cost-free. All figures are averages across 41 customer profiles.

Model	Info. Gathering Cost	Total Cost	Info. Gathering Time	Total Time
<i>Human baseline</i>				
Human (5 min limit)	—	\$5.43	—	6.0 min
Human (30 min limit)	—	\$14.04	—	15.6 min
<i>AI models (all tools)</i>				
Claude Sonnet 4	\$0.66	\$1.74	1.5 min	2.7 min
Grok 4	\$0.21	\$1.29	2.6 min	3.8 min
Gemini 2.5 Pro	\$0.10	\$1.18	1.3 min	2.5 min
GLM 4.6	\$0.12	\$1.20	1.7 min	2.9 min
MiniMax M2	\$0.06	\$1.14	2.2 min	3.4 min
Average (all tools)	\$0.23	\$1.31	1.9 min	3.1 min
<i>AI models (web-only)</i>				
Claude Sonnet 4	\$0.68	\$1.76	1.0 min	2.2 min
Grok 4	\$0.26	\$1.33	2.7 min	3.9 min
Gemini 2.5 Pro	\$0.11	\$1.19	1.3 min	2.5 min
GLM 4.6	\$0.19	\$1.27	1.4 min	2.6 min
MiniMax M2	\$0.12	\$1.20	2.5 min	3.7 min
Average (web-only)	\$0.27	\$1.35	1.8 min	3.0 min

indexed by Europe PMC or ORCID, which cover primarily academic publications. For most profiles in our dataset, this substitution had little effect, as publication databases contained comparable evidence. However, industry customers were specifically selected based on non-publication work outputs, so tool access steered models away from the most relevant sources for these customers.

3.2 Cost Comparison

All AI models produced customer reports faster and more cheaply than either human baseline. Comparing total costs—including human review of AI-generated reports—the most expensive model tested (Claude Sonnet 4, \$1.74 total) was approximately 8 times cheaper than fully manual screening (\$14.04), while the cheapest configuration (MiniMax M2 with tools, \$1.14 total) was approximately 12 times cheaper. The cost reductions were more dramatic when comparing information gathering alone: AI models averaged \$0.23–\$0.27 per customer, roughly 50–60 times cheaper than the human baseline.

Across AI models, cost per customer did not correlate with overall pass rates ($R^2 = 0.01$). The best-performing model, Gemini 2.5 Pro with all tools, was also the second cheapest configuration. While the open-source models (GLM 4.6 and MiniMax M2) had per-token prices 2–8× lower than Gemini 2.5 Pro, additional web search queries and higher token consumption (often from extra tool calls) eliminated any meaningful cost advantage.

Even with substantially cheaper AI models, per-customer costs were dominated by web search queries. For the model with the lowest per-token price (MiniMax M2), web search accounted for 60% of total processing cost. Alternative web search providers—including custom tools offered by some model developers—could

reduce costs further, but the lack of standardization across such tools would require adapting prompts and source extraction for each provider.

3.3 Flag Error Analysis

We manually reviewed a random sample of 126 flag errors across AI and human screeners and assigned each to one of five categories:

- **Criterion Deviation:** The screener's flag determination conflicted with the criteria used to establish ground truth. These criteria were often specified explicitly in the screening prompt, though some recurring edge cases were not addressed directly and required interpretation.
- **Search Failure:** The screener did not locate information that was central to the flag determination but that other screeners successfully found.
- **Missing Response:** The screener produced no response in the relevant field, or the response was incomplete or not formatted according to task instructions.
- **Ambiguous Case:** The screener had access to all relevant information, but no clear rule in the flag determination criteria resolved the case, requiring a judgment call where reasonable screeners might disagree.
- **Source Misinterpretation:** The screener cited appropriate sources but misrepresented their content, leading to an incorrect flag.

3.3.1 Error Distribution

Table 3 summarizes error category distributions across screener types.

Except for errors in the search failure category, the frequency of each error type was similar across screeners. Criterion deviation was the most common category, accounting for 39.7% of all errors. On inspection, a small number of recurring patterns accounted for the majority of mistakes in this category, such as failing to flag personal (non-institutional) email addresses used by otherwise reputable researchers in China.

Because grading flags required substantial manual work, we only graded them after settling on the final prompts provided to human and AI models. We expect that including explicit guidance addressing these recurring cases could substantially reduce criterion deviation errors for both AI and human screeners.

Search failure was the next most common error type. Notably, web-only models had a lower rate (16.7%) than AI models with access to specialized tools or human screeners. Although we did not examine this pattern in detail, it resembles what we observed for work relevance: models with additional tools performed fewer web searches, potentially missing information available only through general web results.

Missing responses accounted for 21.4% of errors. For human screeners, these typically arose from deliberate decisions to skip a criterion—for example, skipping remaining checks after confirming a customer's institution appears on a sanctions list—that coincided with a flag error. For AI models, missing responses often resulted from outputs that did not adhere to the requested table format or, occasionally, from extraction failures when parsing model responses.

Ambiguous cases and source misinterpretation were the least common, accounting for 11.1% and 0.8% of errors respectively. Ambiguous cases typically involved conflicting or partial evidence that made it difficult to establish clear ground truth. Source misinterpretation was rare despite the fact that around 15%

Table 3. Distribution of flag errors by category and screener type. Percentages are computed within each screener type (column), showing what share of that screener type’s errors fell into each category. Counts are in parentheses; the 126 total errors were sampled across all screeners. “Human Baseline” combines both the 5-minute and 30-minute snapshots. Examples illustrate representative instances; error categories are defined in the text.

Error Category	Example	Human Baseline	AI (All Tools)	AI (Web Only)	Overall
Criterion Deviation	Flagging a biology dept. as “not in life sciences” because its university appears on a sanctions list	35% (6)	39% (19)	42% (25)	40% (50)
Search Failure	Missing that a researcher moved to a different institution	35% (6)	35% (17)	17% (10)	26% (33)
Missing Response	Empty response field or abruptly terminated output	18% (3)	20% (10)	23% (14)	21% (27)
Ambiguous Case	Email domain matches recently outdated company domain	12% (2)	6% (3)	15% (9)	11% (14)
Source Misinterp.	Assuming connections between nearby organizations without evidence	0% (0)	0% (0)	2% (1)	1% (1)

of screener responses contained some factual inaccuracy, as measured by the source fidelity metric. This suggests that most of these inaccuracies were minor and did not affect the final flag determination.

3.3.2 Geographic Variation

Error rates varied by customer region (Figure 5). European customers had the highest pass rates, potentially reflecting better documentation in the sources most available to screeners (English-language web search results and EuropePMC publications). Chinese customers showed higher missed flag rates on domain verification—15.0% compared to under 5% for other regions—as screeners often did not flag otherwise reputable researchers who used personal (non-institutional) email addresses. This aligns with the criterion deviation pattern noted above: the screening prompt did not explicitly address cases where personal email use is a professional norm rather than a verification concern.

Chinese customers also showed elevated undetermined rates on sanctions screening (7.5%), driven by institutions that appear on some countries’ sanctions lists but not others, creating genuine ambiguity about whether a flag is warranted. US customers showed unexpectedly high affiliation verification errors (10.0% missed flags, 10.0% false flags), but we did not identify any consistent pattern when reviewing the full response transcripts, and the small sample size (6 profiles) limits the conclusions that can be drawn from this region.

4 DISCUSSION

4.1 Key Findings

We evaluated AI models on follow-up screening of customers who ordered sequences of concern, comparing models with web search alone against those with additional specialized tools (Consolidated Screening List API, Europe PMC, ORCID), and against a human baseline. While our evaluation took place outside a DNA synthesis company, we simulated realistic screening scenarios with customers who have verifiable research backgrounds. These results warrant piloting AI-aided legitimacy screening at DNA synthesis companies and possibly other providers with similar risk profiles (e.g., cloud labs and pathogen repositories). This aligns with existing screening guidelines that pair sequence screening with legitimacy screening and recommend follow-up when orders are flagged (U.S. Department of Health and Human Services, 2023; International Gene Synthesis Consortium, 2024).

AI models outperformed the 30-minute human baseline on most metrics, with the best model (Gemini 2.5 Pro with tools) achieving an 89.8% pass rate compared to 79.5% for humans. However, we invested substantially more effort in prompt development than in training human screeners. Experienced screeners with domain expertise might outperform current models on ambiguous cases requiring contextual judgment—cases our evaluation included but may have undersampled. This performance pattern reflects how LLMs are trained: instruction-tuned models excel at following written procedures and structured checklists, but they do not learn from experience during deployment unless explicitly updated through fine-tuning. Our approach, encoding screening guidance in detailed prompts with explicit rules for acceptable sources, plays to these strengths while avoiding the need for per-case adaptation.

Models showed greater variation across source quality metrics than on flag accuracy. Flag accuracy pass rates clustered within a narrow range for both AI and human screeners. However, source quality and fidelity metrics showed wider spreads: Claude Sonnet 4 struggled with institution type verification sources where the human baseline excelled; conversely, Grok 4 with tools substantially outperformed humans on background work relevance.

AI models completed information gathering in under 2 minutes on average compared to roughly 15 minutes for the human baseline—approximately an 8-fold throughput advantage. This speed becomes increasingly important as the synthetic nucleic acid market grows at a projected compound annual growth rate of ~15% (Fady et al., 2025b). AI-assisted screening enables providers to scale verification capacity without proportional increases in human labor.

Access to specialized tools provided modest but consistent improvements across most tasks. The advantage was most pronounced for sanctions screening, where models with Consolidated Screening List API access substantially outperformed web-only configurations. Sanctions lists are typically available only through APIs or downloadable files, not as indexed webpages, making direct API access particularly valuable. However, tool access reduced performance on background work search by steering models away from web searches that would find patents and news articles not indexed in academic databases. Our implementation queried only the U.S. Consolidated Screening List; integrating additional sanctions databases (EU lists, OFAC, UN Security Council) could further improve coverage.

4.2 Implications for Screening Practice

The mix of tasks performed here could plausibly be part of one legitimacy screening instance, but they could also be split into onboarding tasks and follow-up tasks. Crucially, the ship/follow-up/reject decision

should remain with humans. Fully autonomous AI screening risks introducing biases from language models to order decisions, and should be avoided until we have stronger performance guarantees. AI error rates, while lower than our human baseline on these information-gathering tasks, remain non-negligible for security-critical applications where false clearances could enable misuse.

Cost reduction is the primary driver for piloting AI-assisted screening. The best-performing model (Gemini 2.5 Pro with tools) cost \$1.18 per customer including human review, compared to \$14.04 for manual screening—roughly a 10-fold reduction. Information-gathering tasks alone cost \$0.23 per customer on average for AI models, roughly 50 times cheaper than the human baseline. Open-source tools like Cliver (Acelas, 2025) could accelerate adoption by lowering barriers for providers who already screen and reducing costs for those considering implementation.

Onboarding tasks typically include name checks against watchlists, verification of institutional affiliation and email domain, and confirmation of research purpose. Follow-up tasks, triggered when orders contain sequences of concern, aim to collect information about whether the customer has a legitimate need for the sequence. Follow-up often requires requesting additional documentation from customers (biosafety approvals, grant awards, publications) and then verifying it. AI excels at the verification component but is yet to be tested on tasks requiring human interaction to request additional information.

AI-assisted legitimacy screening sits within a broader ecosystem of measures for enhancing biosecurity in synthetic nucleic acid access. It can be strengthened by frontloading information collection. Standardized templates, such as those developed by IBBIS for customers ordering sequences of concern (International Biosecurity and Biosafety Initiative for Science, 2025), prompt customers to provide verification materials upfront. Whitelisting verified customers for specific sequences with periodic re-verification (Crawford et al., 2024) similarly converts requests into verification tasks. At present, the value of AI-assisted screening lies in accelerating these verification tasks that currently bottleneck workflows, while humans retain responsibility for requesting additional information and making final fulfillment decisions.

Emerging EU regulation is supportive of this approach. The proposed European Biotech Act (COM(2025) 1022), published in December 2025, introduces harmonized EU-wide biosecurity obligations for “biotechnology products of concern,” requiring providers to assess the legitimate need of prospective customers through screening procedures explicitly likened to know-your-customer assessments. We believe that the integration of AI tools into KYC checks as described here would not classify as a high-risk AI system under the EU AI Act (Regulation (EU) 2024/1689). Providing synthetic nucleic acids does not fall within any of the Annex III high-risk categories. Further, the system is explicitly designed as human-in-the-loop: the final ship, follow-up, or reject decision remains with human operators, meaning the AI does not autonomously determine outcomes that affect individuals’ rights. Either condition alone would suffice to exempt the system from high-risk classification under Article 6.

4.3 Limitations

Our screening approach depends on publicly available information about the customer. This affects both AI and human screeners and is inherent to any information-gathering system. However, dependence on public documentation may introduce bias: well-documented customers receive faster verification, while customers with limited online presence—including researchers at institutions with minimal web visibility, early-career personnel, and stealth startups—face longer review times or heightened scrutiny. This could disproportionately affect researchers from low- and middle-income countries, who already face barriers to accessing biotechnology.

Our evaluation dataset likely overrepresents well-documented researchers. By design, all profiles have discoverable publications, patents, or institutional affiliations—information necessary both for constructing realistic profiles and for screeners to verify them. This overrepresents established researchers relative to typical DNA synthesis customers, which include laboratory technicians, students, and industry personnel with minimal publication records. Performance on our benchmark may overestimate real-world accuracy for these less-documented customers, who will remain comparatively more reliant on direct communication and manual verification.

5 CONCLUSION

This work builds on years of guidance on DNA synthesis legitimacy screening (Carter and Friedman, 2015; Alexanian and Carter, 2024; International Gene Synthesis Consortium, 2024). We demonstrated that AI-assisted legitimacy screening is substantially less costly and faster than human-only screening, while achieving comparable or higher accuracy on information-checking tasks. These results support leveraging AI-assisted legitimacy screening as part of industry-wide biosecurity. Although we conducted this study with the DNA synthesis industry in mind, our results broadly support the use of AI-assisted legitimacy screening for any dual-use product in biotech, including live viral or microbial cells, equipment, and other materials of concern. Future work should explore hybrid AI–human systems where models flag uncertain cases for human review, fine-tuned on provider-specific data, and integrated with broader customer onboarding workflows.

DATA AVAILABILITY STATEMENT

The prompts, evaluation framework, and open-source tool (Cliver) developed for this work are available on GitHub at <https://github.com/alejandroAcelas/cliver>.

AUTHOR CONTRIBUTIONS

AA designed the study, developed the evaluation framework, collected data, conducted experiments, and wrote the manuscript. KF and HP served as human baseline screeners and provided critical feedback on the manuscript. PEF and CN provided strategic guidance and contributed to the writing of the discussion and regulatory analysis sections. All authors reviewed and approved the final manuscript.

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CONFLICT OF INTEREST

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

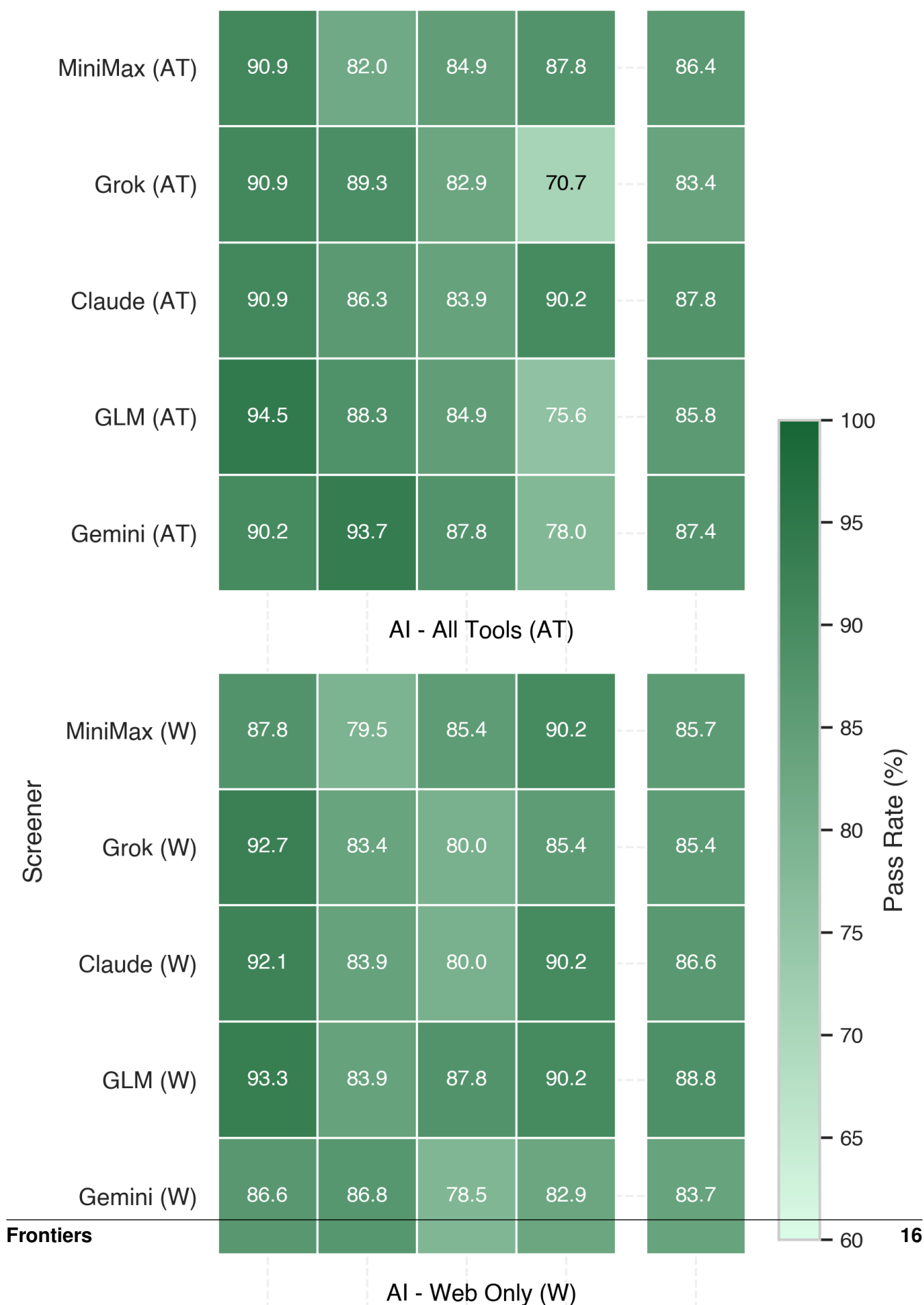
SUPPLEMENTARY MATERIAL

427 The Supplementary Material for this article can be found online at: [link to be provided by journal]

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Pass Rate by Screener and Test Category



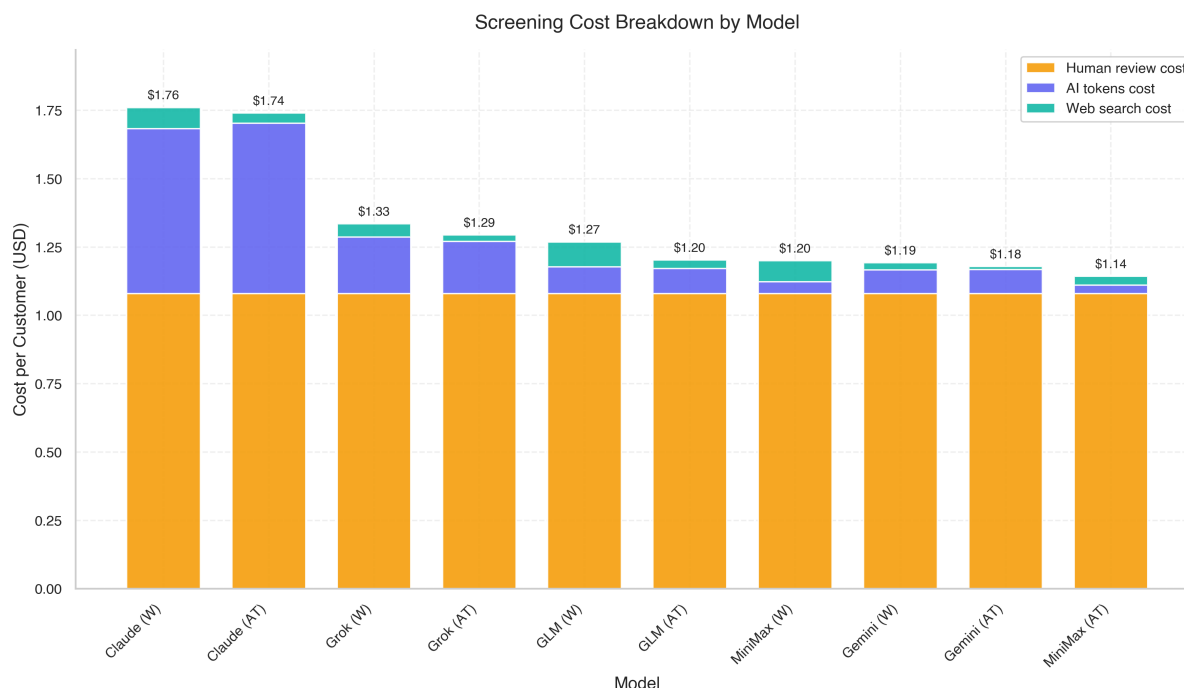


Figure 3. Per-customer screening cost breakdown by model configuration. Stacked bars show three cost components: human review of the AI-generated report (orange), AI token costs for input and output (purple), and web search queries via Tavily (teal). Human review cost was estimated only for the best performing model configuration (Gemini 2.5 Pro with all tools), and it is taken as constant across configurations ($\sim \$1.08$). All values are averages across 41 customer profiles.

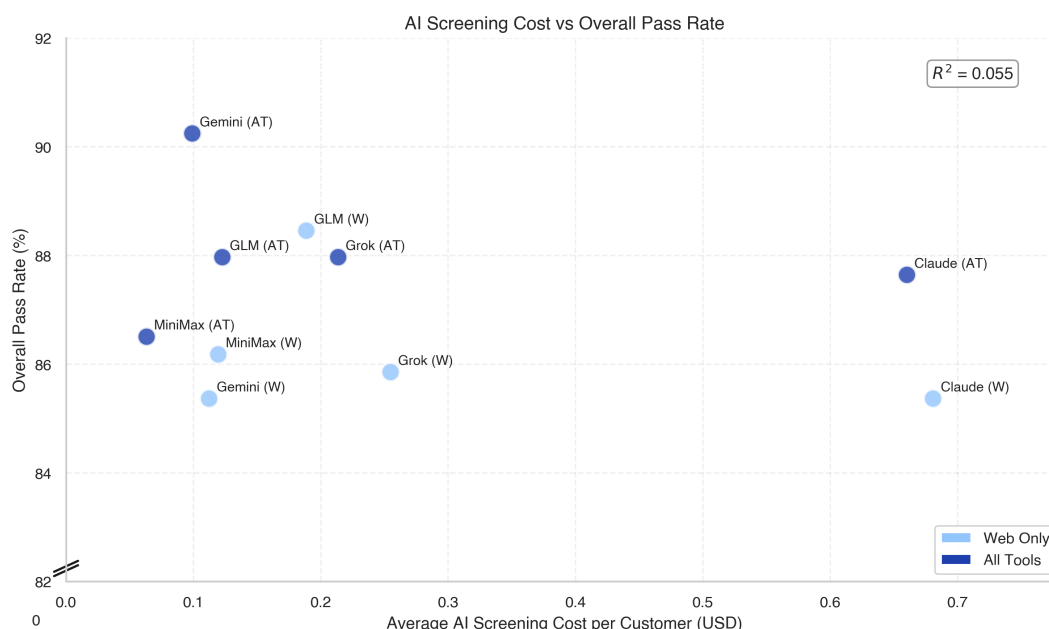


Figure 4. AI screening cost versus overall pass rate. Each point represents one model configuration (five models $\times$ two tool conditions). The x-axis shows average AI information-gathering cost per customer (excluding human review); the y-axis shows the overall pass rate across all four metrics (note the broken y-axis starting at $\sim 83\%$). All values are averages across 41 customer profiles.

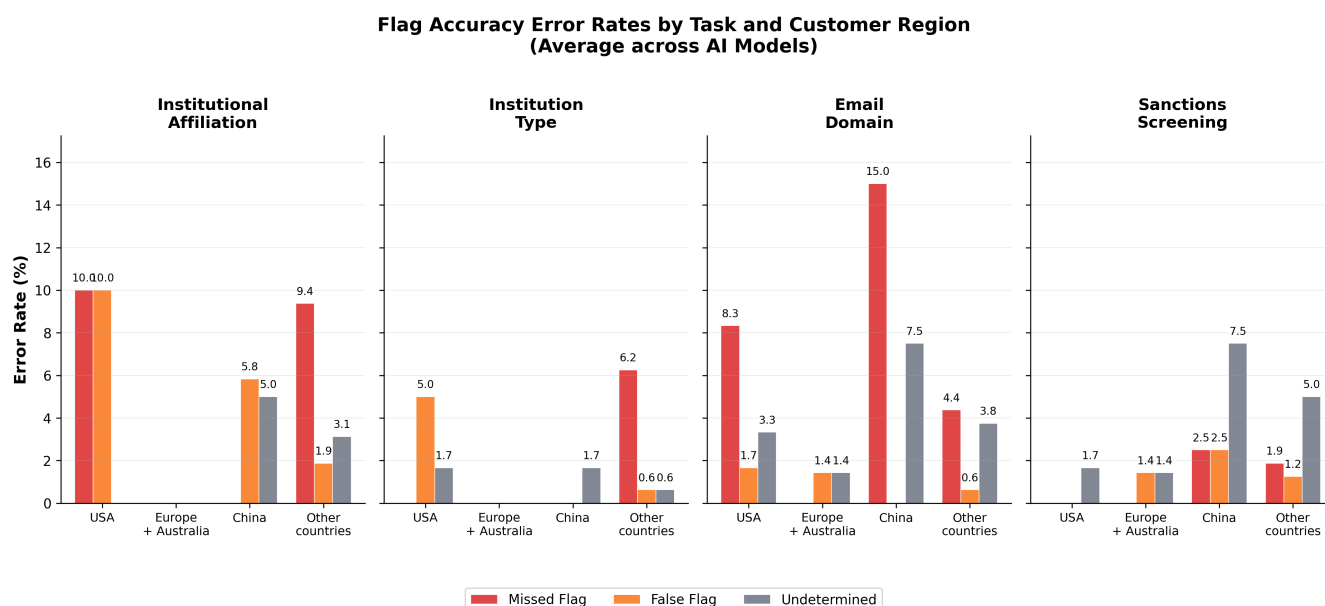


Figure 5. Flag accuracy error rates by task and customer region, averaged across all AI model configurations. Each panel shows one verification task; bars indicate the rate of missed flags (red; screener reported no flag when ground truth was flag), false flags (orange; screener reported flag when ground truth was no flag), and undetermined responses (gray; screener reported undetermined when ground truth was either flag or no flag). Regions are defined by institution location: US (6 profiles), Europe + Australia (7 profiles), China (12 profiles), and Other countries (16 profiles). Error rates are computed as a percentage of all screener–profile pairs within each region.